

B. LIPIKA

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Profile Summary

AI/ML Engineer focused on industrial AI, time-series forecasting, and agentic AI systems. Built sensor-data analytics pipelines and LangGraph-based agents using Python, SQL, PyTorch, FastAPI, and local LLMs, with hands-on work in forecasting, anomaly detection, and evidence-driven AI workflows. Designs systems that keep numerical computation deterministic and use LLMs for reasoning and explanation. Indian utility patent holder with experience across ML, data engineering, and embedded IoT.

Technical Skills

Programming	Python, SQL, Java
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, LSTM, SHAP
Time-Series Analytics	Forecasting, Trend Analysis, Seasonality, Stationarity, Anomaly Detection, Changepoint Detection, Rolling Statistics
Generative AI	LangGraph, LangChain, Ollama, Local LLMs, NLP, Tool Calling, MCP, AI Agents
LLM Technologies	Llama, Qwen, Phi, Local LLM Inference, LLM Evaluation and Benchmarking
Agent Orchestration	LangGraph-based Routing, Tool Execution, Stateful Workflows, Model-Swappable Agents, Evidence-Driven Reasoning
Data Engineering	Pandas, NumPy, Parquet, DuckDB, PyArrow, Data Cleaning, Data Profiling, Chunk-Based Processing
Backend / Tools	FastAPI, Flask, REST APIs, Git, GitHub, Docker, Jupyter, Postman, WSL, VS Code

Professional Experience

AIML Trainee

Jan 2026 – Present

Tiny Prism Labs, Manipal

- Developing an AI-powered industrial analytics assistant that converts natural-language queries into data retrieval, analysis, and visualization workflows for multivariate sensor time-series data.
- Built a Python query-understanding pipeline to identify equipment, metrics, operations, filters, and time ranges from natural-language requests.
- Developed Flask REST APIs and a schema-aware tool registry for SQL retrieval, dataset inspection, analytical execution, and dynamic tool invocation.
- Implemented reusable time-series analytics for trend, seasonality, stationarity, anomaly, changepoint, and correlation analysis.
- Integrated local LLMs with evidence-driven workflows, separating deterministic numerical computation from model-based reasoning and interpretation.

Projects

Industrial Time-Series Forecasting

[GitHub](#)

Python, PyTorch, IBM Granite TTM-R3, Pandas, FastAPI, Time Series

- Fine-tuned IBM Granite TTM-R3 for **15-minute industrial INFLOW forecasting** using chronological training, validation, and test splits.
- Evaluated long-horizon forecasting using **MAE, RMSE, MAPE, and sMAPE**, with repeat-period baselines for comparative evaluation.
- Performed error analysis across normal operating conditions and extreme inflow spikes to identify forecasting strengths and limitations.
- Built a **FastAPI forecasting service** that maintains rolling forecasts from recent sensor observations and serves stored short-horizon predictions through an API.

Industrial AI Agent Platform

[GitHub](#)

Python, LangGraph, Local LLMs, SQL, Pandas, FastAPI

- Designed a **LangGraph-based orchestration system** for routing industrial queries across analytics, reasoning, and tool-execution workflows.
- Built schema-aware tools for dataset inspection, SQL querying, time-range discovery, and deterministic time-series analysis.
- Designed the architecture to support interchangeable LLMs and extensible tools while keeping analytical computation independent from model-generated reasoning.
- Implemented a modular Router → Analytics → Reasoning → Answer workflow to separate query routing, deterministic analysis, and LLM-based response generation.

Python, Local LLMs, AI Agents, SQL, Pandas, Scikit-learn

- Developed an AI-assisted data-science workflow for automated dataset profiling, schema discovery, analytical workflow generation, and ML experiment support.
- Implemented deterministic analytical tools for trend, seasonality, stationarity, anomaly, changepoint, correlation, and pattern analysis.
- Applied an evidence-driven architecture in which numerical results are generated by Python tools and passed to the reasoning agent for interpretation.

Industrial Vibration Anomaly Detection

GitHub

Python, Pandas, Parquet, Isolation Forest, Ollama

- Built an anomaly-detection workflow for a **2.3M+ row industrial vibration dataset** using statistical analysis and Isolation Forest.
- Performed data-quality analysis across vibration features and metadata, including missing-value, duplicate, and sensor-consistency checks.
- Developed visualization workflows to examine vibration patterns and identify anomalous sensor behavior.
- Integrated local-LLM reasoning with analytical outputs to support interpretation of detected sensor patterns and anomalies.

Industrial Sensor Data Lake and Analytics Pipeline

GitHub

Python, DuckDB, Parquet, SQL, Pandas, PyArrow, LangChain

- Built an end-to-end industrial sensor data pipeline for incremental ingestion, data-quality validation, transformation, and partitioned Parquet storage.
- Implemented **DuckDB-based analytical layers** with hourly and daily rollups for efficient SQL analytics over partitioned sensor datasets.
- Added a LangChain-based natural-language interface that converts user requests into validated SQL queries over the analytical data layer.

Research and Innovation

Retina AI – Visual Data to Tactile and Voice Feedback

Team Lead

Python, YOLOv8, OpenCV, ESP32

- Developed a computer-vision assistive system combining object detection with tactile vibration and voice feedback using embedded hardware.

AgriFlow – IoT Smart Irrigation System

Indian Utility Patent No. 202541061616

ESP8266, IoT, Sensors

- Developed a sensor-driven irrigation system with real-time soil monitoring and automated irrigation control; awarded an Indian utility patent.

Technical Events & Industry Exposure

- **NXP Semiconductors — Time-Series / Edge AI:** Explored sensor analytics and edge-AI applications, including NXP’s eIQ Time Series Studio for real-time sensor insights.
- **electronica India 2026 — Bengaluru:** Attended the electronics and technology industry event covering industrial AI, edge computing, automation, embedded systems, and semiconductor technologies.

Education

Bachelor of Engineering – Information Science and Engineering

Canara Engineering College, Mangalore (VTU)

2022 – 2026
CGPA: 8.25 / 10

Certifications and Achievements

- **Google Data Analytics Professional Certificate** – Coursera, 2025
- **Java Programming** – NPTEL, 2024
- **Indian Utility Patent Holder** – AgriFlow Smart Irrigation System, Patent No. 202541061616
- **2nd Runner-Up** – VTU Mini Project Competition